

Task 1 – AI-Powered Inbound Call Booking Assistant

 **Date** [🔗](#)

Jul 3, 2025

 **Participants** [🔗](#)

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 [Task 1 documentation link](#) [🔗](#)

 **Project Overview** [🔗](#)

Workflow Name: AI-Powered Inbound Call Booking Assistant

Goal: Automatically handle inbound calls, transcribe audio to text, extract booking intents using LLM, create booking in Google Calendar, and log data in Airtable CRM.

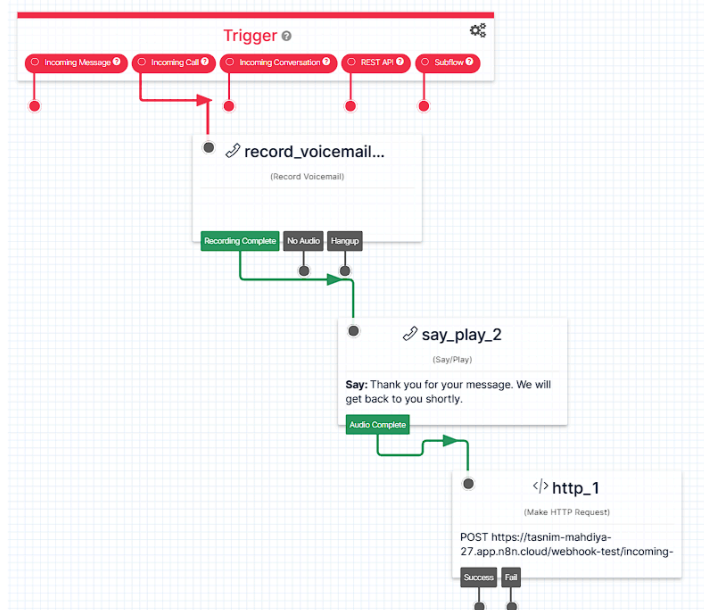
Expected Flow:

Inbound Call → Speech-to-Text (Deepgram) → LLM (DeepSeek) → Booking System (Google Calendar) → CRM System (Airtable)

 **System Architecture** [🔗](#)

```
1 flowchart TD
2   A[Twilio Inbound Call] --> B[Webhook Trigger (n8n)]
3   B --> C[Deepgram Speech-to-Text]
4   C --> D[LLM - DeepSeek AI (Intent Extraction)]
5   D --> E[Code Node - Data Cleanup & Fixes]
6   E --> F[Google Calendar - Create Booking]
7   E --> G[Airtable - Log Entry]
8
```

 **Twilio studio flow** [🔗](#)



Complete Twilio studio flow

The screenshot shows the configuration screen for the **Record Voicemail** widget in Twilio Studio. The screen has a header with a back arrow, the widget name **Record Voicemail**, an information icon, and a **Hide** button. Below the header are two tabs: **Config** (selected) and **Transitions**. The **Config** tab contains the following settings:

- WIDGET NAME ***: `record_voicemail_1`
- STOP RECORDING AFTER**: `5` SECONDS OF SILENCE
- STOP RECORDING ON KEYPRESS**: (Empty text field)
- MAX RECORDING LENGTH**: `45`

At the bottom of the configuration area is a blue **Save** button. To the right of the **Save** button are icons for a document and a trash can.

Call record length and silence

← Make HTTP Request ⓘ Hide

<< Config Transitions

The URL you would like to make a request to.

• Content type

Form URL Encoded

The content type of the request body.

Request body

```
RecordingUrl=
{{widgets.record_voicemail_1.RecordingUrl}}
Caller={{trigger.call.From}}
```

Save

HTTP request body function

Key Node Descriptions [🔗](#)

1. Twilio Webhook (Production) [🔗](#)

- **Type:** HTTP POST
- **Configuration:** Twilio Console > Phone Number > Voice > Webhook > Custom URL (Production URL from n8n)

2. Deepgram Node [🔗](#)

- **Input:** Audio stream from inbound call
- **Output:** English/Malay transcript via `results.channels[0].alternatives[0].transcript`
- **Multilingual Enabled:** Yes

3. LLM Generator Node (DeepSeek) [🔗](#)

Prompt JSON:

```
1 {
2   "model": "deepseek-chat",
3   "messages": [
4     {
5       "role": "system",
6       "content": "You are a helpful booking assistant who understands both Malay and English..."
7     },
8     {
9       "role": "user",
10      "content": "{{ $json.results.channels[0].alternatives[0].transcript }}"
11    }
12  ],
13  "temperature": 0.2
14 }
```

4. Code Node: Date Correction + ISO Time Formatting [🔗](#)

```

1  const raw = $json.choices[0].message.content;
2  const clean = raw.replace(/```json\n?/, '').replace(/```/, '').trim();
3  const parsed = JSON.parse(clean);
4
5  if (parsed.date === "tomorrow") {
6    const tomorrow = new Date();
7    tomorrow.setDate(tomorrow.getDate() + 1);
8    parsed.date = tomorrow.toISOString().split('T')[0];
9  } else {
10   const fixed = new Date(parsed.date);
11   fixed.setFullYear(2025);
12   parsed.date = fixed.toISOString().split('T')[0];
13 }
14
15 const localDateTime = new Date(`${parsed.date}T${parsed.time || '10:00'}:00+08:00`);
16 parsed.formattedDate = localDateTime.toISOString();
17
18 return {
19   json: parsed
20 };
21

```

5. Google Calendar Node [🔗](#)

- **Start:** `{{ $json.formattedDate }}`
- **End:** `{{ new Date(new Date($json.formattedDate).getTime() + 60*60*1000).toISOString() }}`
- **Description:** Dynamic content based on AI parsing
- **Time Zone:** UTC+8 (Malaysia)

⚠️ Note on Time Behavior:

Although the AI extracts and formats the booking time correctly in Malaysia Standard Time (UTC+8), Google Calendar may display it with timezone offset adjustments in the backend (e.g., shifting to UTC time). This does **not** affect the actual booked time shown on the calendar front-end, which remains accurate for Malaysian users. The logic ensures correct day and hour despite backend timezone conversions.

6. Airtable CRM [🔗](#)

- Logs:
 - Customer (Phone)
 - Date & Time (Booking)
 - Branch (Optional)
 - AI Summary

♦ Limitations [🔗](#)

Type	Description
------	-------------

❌ Voice AI Reply	AI does not speak back to the caller due to response complexity
🕒 Limited Call Time	Call ends after ~1 minute due to Deepgram audio limits
? No Response-to-Webhook	Not implemented for Twilio verbal reply
❌ Incorrect Year from LLM	Code node corrects 2024 to 2025
🕒 UTC Offset Visible	Google Calendar UTC+8 shows +1 day in backend

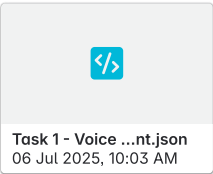
🌟 Highlights [🔗](#)

- Humanlike conversation from transcript using DeepSeek
- Multilingual: Malay + English auto-detection
- Accurate calendar booking via time zone correction
- Airtable integration for complete CRM visibility

📖 Future Improvements [🔗](#)

- Add **Text-to-Speech** response to users
- Implement **Webhook response** to speak via Twilio
- Enable **Session History Logs** with richer detail
- Allow **Dynamic conversation** with booking clarification steps

📌 n8n Workflow: [🔗](#)



Output Example [🔗](#)

📌 Google Calendar Booking [🔗](#)

```
1 Booking intent from AI assistant.
2 Date: 2025-07-06
3 Time: 15:00
4 Branch: `kuala Lumpur
```

AI Bookings

DataAutomationsInterfacesForms

All changes saved

Bookings

+ Add or import

Grid view

Hide fieldsFilterGroupSortColor

+ Create new...

Find a view

	A Caller	formattedDate	A Branch	Transcript	A Intent
13	+601169378360	7/6/2025 3:00pm +08	Kuala Lumpur	hey i want to make a booki...	booking

Airtable record of the booking

AI Booking: Dialysis Session

July 6, 2025, 11:00pm – July 7, 2025, 12:00am

Take meeting notes

Start a new document to capture notes

Booking intent from AI assistant.

Date: 2025-07-06

Time: 15:00

Branch: Kuala Lumpur

10 minutes before

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AI Booking: Dialysis Ses

11pm – 12am

Google Calendar booking event